Tugas IF3051 Strategi Algoritma

Algoritma Pencocokan String dengan Metode Brute Force, Knuth Morris Pratt, dan Boyer-Moore

Oleh: Jonathan Marcel T (13507072)

1. Source Code Program

Class pmatch.java

```
package tugaskecil3;
public class pmatch {
    public int count;
    public int count_bm;
    public int count_kmp;

    //algoritma bruteforce
    public int strmatch_brute(String t,String p){
        int n = t.length();
        int m = p.length();
        int j;
        count = 0;
        for (int i = 0;i <= n-m;i++){
            j = 0;
            count +=1;
            while ((j<m) && (t.charAt(i+j) == p.charAt(j))){
                count +=1;
                j+=1;
                if (j == m)return i;else count+=1;
            }
        }
        return -1;
    }

    //algoritma boyermoore
    public int strmatch_boyermoore(String t,String p){
        int last[] = last_function(p);
        int n = t.length();
        int m = p.length();
        int i = m-1;
        if (i > n-1)return -1;
        int j = m-1;
        count_bm = 0;
        while (i<= n-1)
        {   count_bm +=1;
            if(p.charAt(j)==t.charAt(i)){
                if (j==0)return i;
                else{
                    i-=1;
                    j-=1;
                }
            }
```

```
            else{
                int lo = last[t.charAt(i)];
                i = i+m - Math.min(j,1+lo);
                j = m - 1;
            }

        }
        return -1;
    }

    //fungsi last occurence
    public static int[] last_function(String p){
        int last[] = new int[128];
        for(int i=0;i<128;i++){
            last[i]=-1;
        }
        for(int i=0;i<p.length();i++){
            last[p.charAt(i)]=i;
        }
        return last;
    }

    //algoritma knuth morris pratt
    public int strmatch_kmp(String t,String p){
        int n = t.length();
        int m = p.length();
        int fail[] = fail_function(p);
        int i = 0;
        int j = 0;
        count_kmp = 0;
        while (i<n){
            count_kmp +=1;
            if(p.charAt(j)==t.charAt(i)){
                if(j==m-1)
                    return i-m+1;
                    i+=1;
                    j+=1;

            }
            else if(j>0)j = fail[j-1];
            else i++;

        }
        return -1;
    }

    //fungsi failure
    public static int[] fail_function(String p){
        int fail[] = new int[p.length()];
        fail[0] = 0;
        int m = p.length();
        int j = 0;
        int i = 0;
        while (i<m){
            if(p.charAt(j)==p.charAt(i)){
                fail[i]=j+1;
```

```
                i+=1;
                j+=1;
            }
            else if(j>0)j=fail[j-1];
            else{
                fail[i]=0;
                i+=1;
            }
         }
        return fail;
    }
}
```

Class file reader.java

```
package tugaskecil3;
import javax.swing.JFileChooser;
import java.io.File;
public class filereader {
    private static JFileChooser a = null;
    private String dir;
    private boolean isfile = false;
public filereader(){
    if (a == null)a = new JFileChooser();
    int filestate = a.showOpenDialog(null);
    File f = a.getSelectedFile();
    if(f != null && filestate ==
a.APPROVE_OPTION)setfilepath(f.getAbsolutePath());
    else if (filestate == a.CANCEL_OPTION)isfile = false;
}
public void setfilepath(String path){
    dir = path;
    isfile = true;
}
public String getfilepath(){
    return dir;
}
}
```

Class NewJPanel.java

```
package tugaskecil3;

import java.io.BufferedInputStream;
import java.io.DataInputStream;
import java.io.File;
import java.io.FileInputStream;
import java.io.IOException;

public class NewJPanel extends javax.swing.JPanel {

    /** Creates new form NewJPanel */
    public NewJPanel() {
        initComponents();
    }
private void jButton1ActionPerformed(java.awt.event.ActionEvent evt)
pmatch a = new pmatch();
```

```
        int b=a.strmatch_brute(jTextField1.getText(), jTextField2.getText());
        if (b==-1)jTextField3.setText("tidak ditemukan");
        else {
            jTextField3.setText("ditemukan pada posisi ke "+b+" dari awal
teks");
            jTextField4.setText("jumlah perbandingan karakter yang dilakukan
ialah "+a.count);
        }
}
    private void jButton2ActionPerformed(java.awt.event.ActionEvent evt) {
        pmatch a = new pmatch();
        int b=a.strmatch_kmp(jTextField1.getText(), jTextField2.getText());
        if (b==-1)jTextField3.setText("tidak ditemukan");
        else {jTextField3.setText("ditemukan pada posisi ke "+b+" dari awal
teks");
        jTextField4.setText("jumlah perbandingan karakter yang dilakukan
ialah "+a.count_kmp);
        }
    }
    private void jButton3ActionPerformed(java.awt.event.ActionEvent evt) {
       pmatch a = new pmatch();
        int b=a.strmatch_boyermoore(jTextField1.getText(),
jTextField2.getText());
        if (b==-1)jTextField3.setText("tidak ditemukan");
        else {jTextField3.setText("ditemukan pada posisi ke "+b+" dari awal
teks");
        jTextField4.setText("jumlah perbandingan karakter yang dilakukan
ialah "+a.count_bm);
        }
    }    private void jButton4ActionPerformed(java.awt.event.ActionEvent evt)
{
       DataInputStream dis = null;
        String record = null;
        String buffer = new String();
        filereader freader = new filereader();
        String text = "";
        try {
            File f = new File(freader.getfilepath());
            FileInputStream fis = new FileInputStream(f);
            BufferedInputStream bis = new BufferedInputStream(fis);
            dis = new DataInputStream(bis);

            while ( (record=dis.readLine()) != null ) {
                text += "\n"+record;
            }

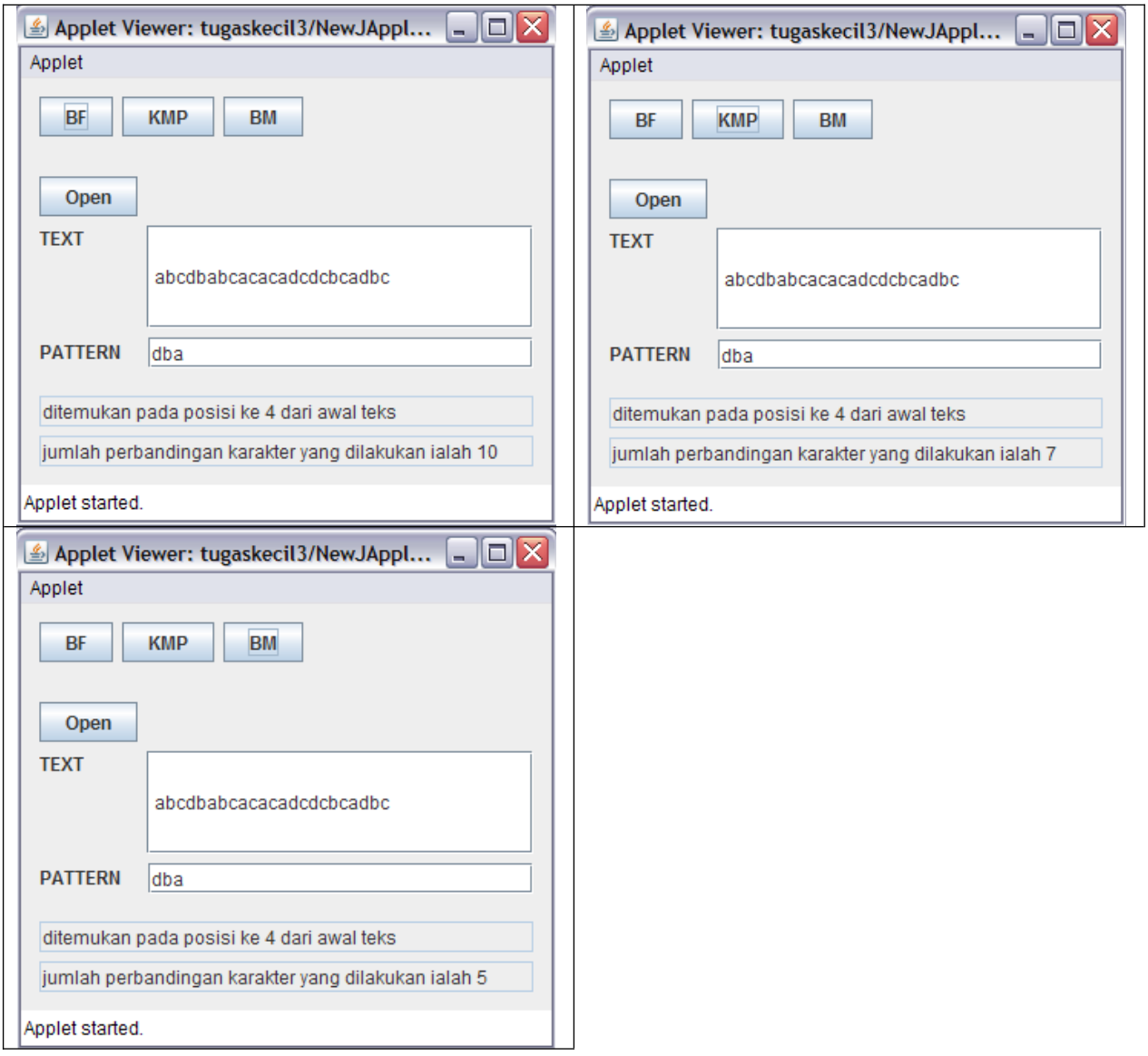
        } catch (IOException e) {

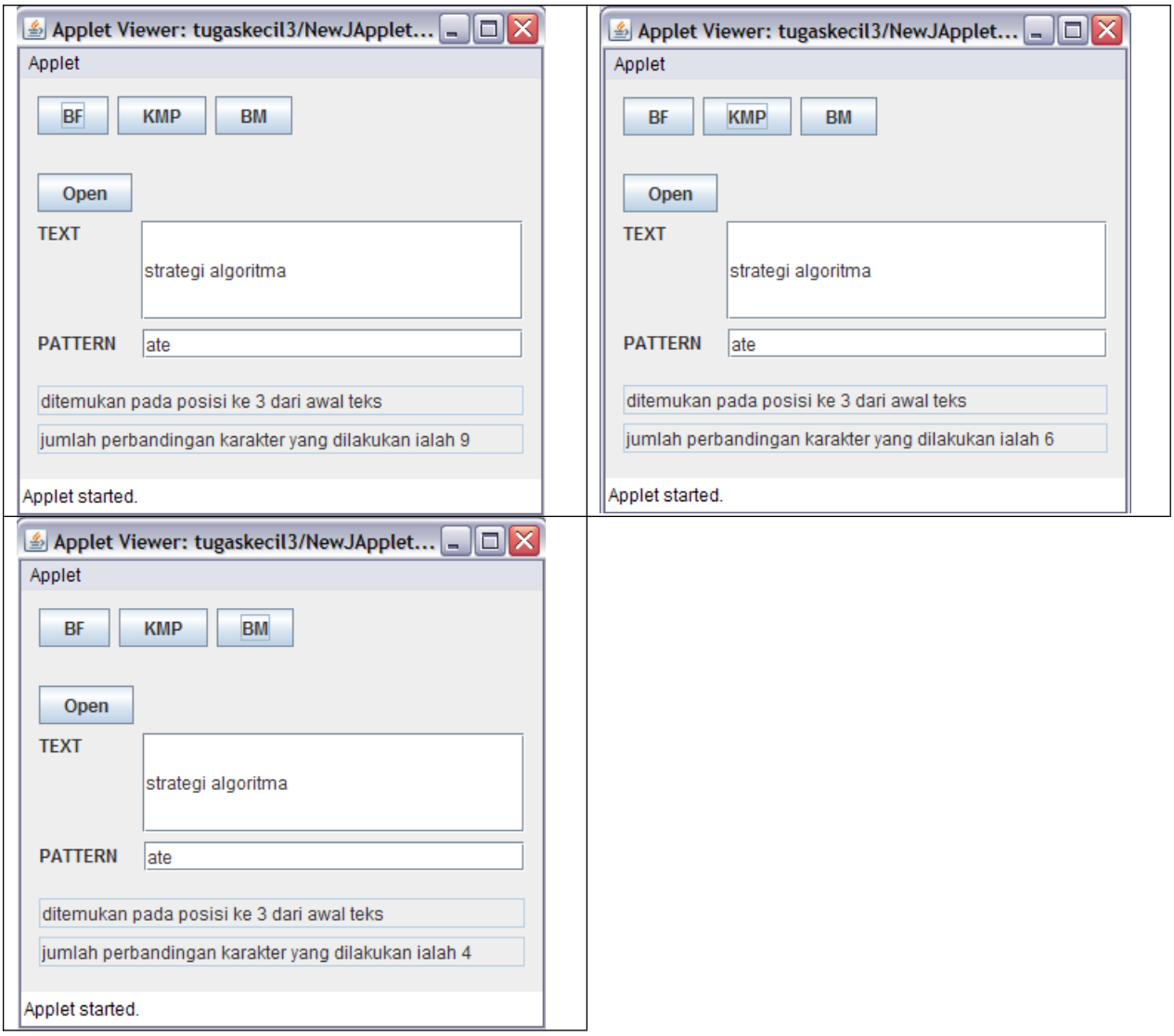
           System.out.println("Uh oh, got an IOException error!" +
e.getMessage());

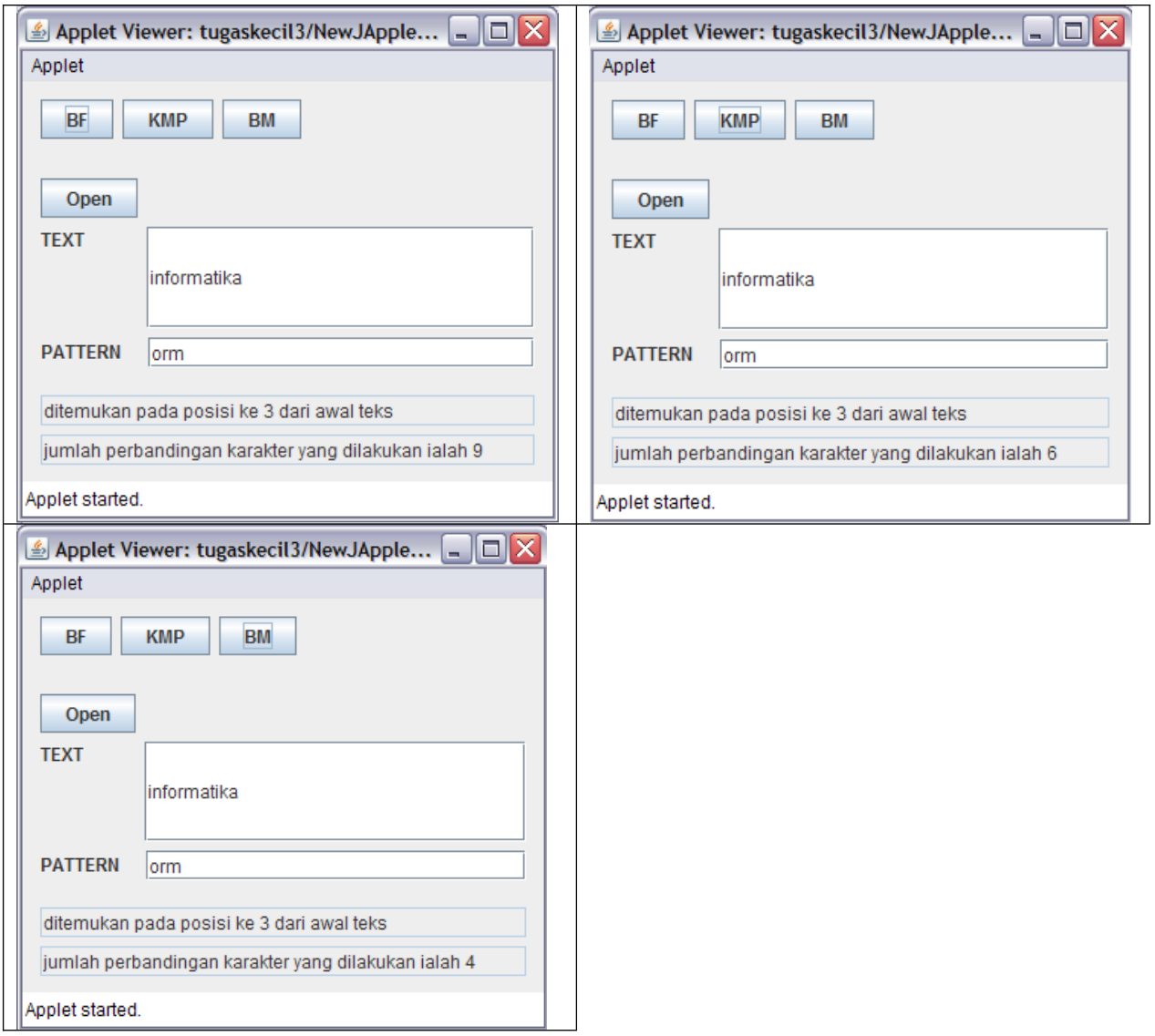
        } finally {
             if (dis != null) {
                try {
                    dis.close();
                } catch (IOException ioe) {
```

```
                }
            }
            jTextField1.setText(text);
        }
    }
```

2. Contoh Eksekusi Program




Applet Viewer: tugaskecil3/NewJApplet...
Applet
BF
KMP
BM
Open
TEXT
strategi algoritma
PATTERN
ate
ditemukan pada posisi ke 3 dari awal teks
jumlah perbandingan karakter yang dilakukan ialah 9
Applet started.
Applet Viewer: tugaskecil3/NewJApplet...
Applet
BF
KMP
BM
Open
TEXT
strategi algoritma
PATTERN
ate
ditemukan pada posisi ke 3 dari awal teks
jumlah perbandingan karakter yang dilakukan ialah 6
Applet started.
Applet Viewer: tugaskecil3/NewJApplet...
Applet
BF
KMP
BM
Open
TEXT
strategi algoritma
PATTERN
ate
ditemukan pada posisi ke 3 dari awal teks
jumlah perbandingan karakter yang dilakukan ialah 4
Applet started.


Applet Viewer: tugaskecil3/NewJApple...
Applet
BF
KMP
BM
Open
TEXT
informatika
PATTERN
orm
ditemukan pada posisi ke 3 dari awal teks
jumlah perbandingan karakter yang dilakukan ialah 9
Applet started.
Applet Viewer: tugaskecil3/NewJApple...
Applet
BF
KMP
BM
Open
TEXT
informatika
PATTERN
orm
ditemukan pada posisi ke 3 dari awal teks
jumlah perbandingan karakter yang dilakukan ialah 6
Applet started.
Applet Viewer: tugaskecil3/NewJApple...
Applet
BF
KMP
BM
Open
TEXT
informatika
PATTERN
orm
ditemukan pada posisi ke 3 dari awal teks
jumlah perbandingan karakter yang dilakukan ialah 4
Applet started.